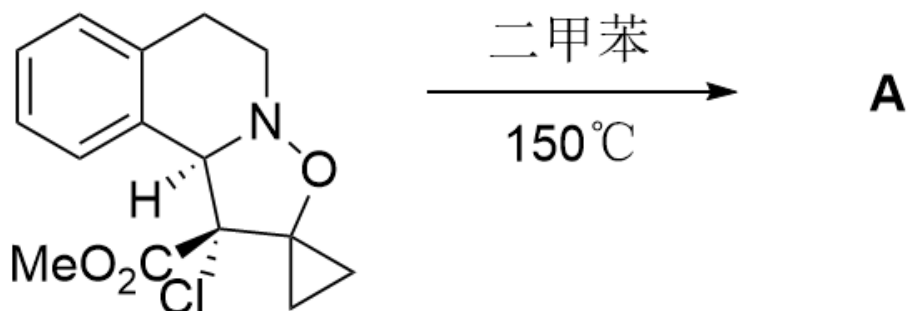


Question

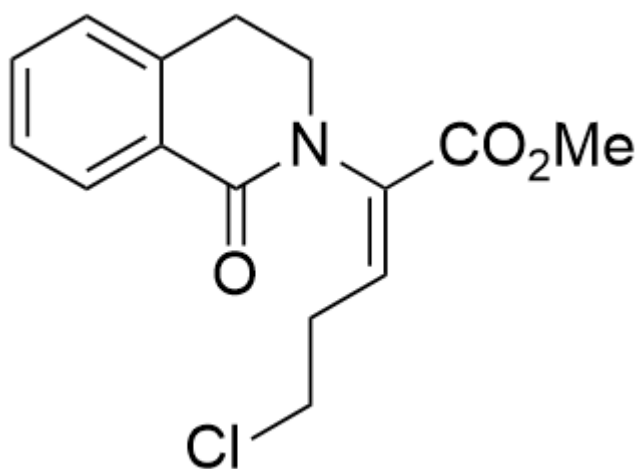


[H][C@]12C3=CC=CC=C3CCN1OC4(CC4)[C@]2(C(OC)=O)Cl>Xylene,150°C>[A], **A** is the product

Given that reaction product **A** contains an amide bond, and the first step of the reaction is a pericyclic reaction, without considering enantiomers, provide the structural formula of reaction product **A**.

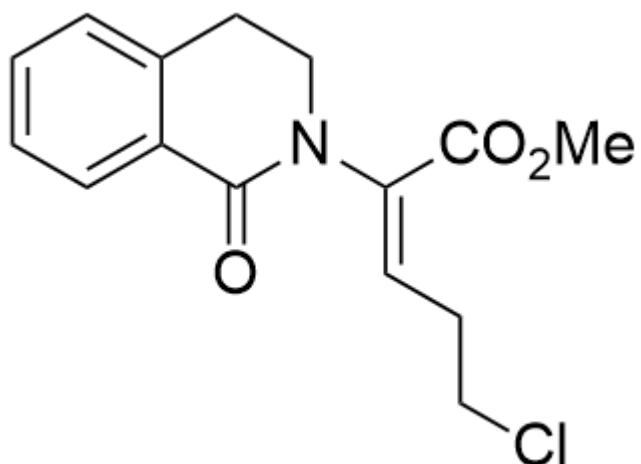
A. All other options are incorrect

B.



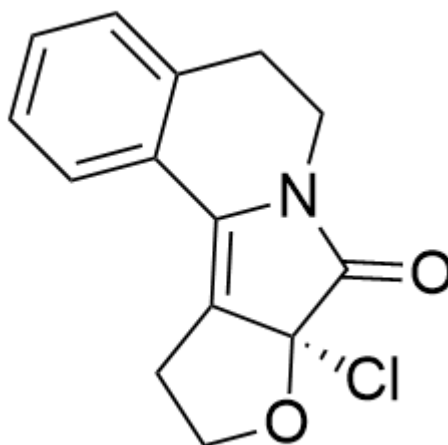
O=C1C2=CC=CC=C2CCN1/C(C(OC)=O)=C\CCCl

C.



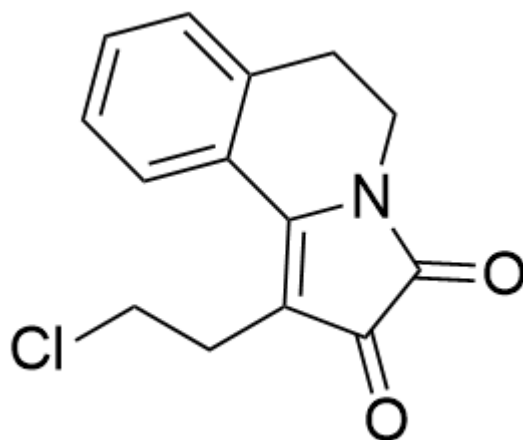
O=C1C2=CC=CC=C2CCN1/C(C(OC)=O)=C/CCCl

D.



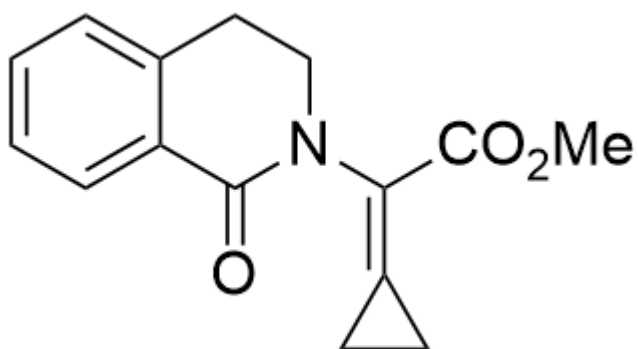
O=C1[C@]2(Cl)C(CCO2)=C3C4=CC=CC=C4CCN31

E.



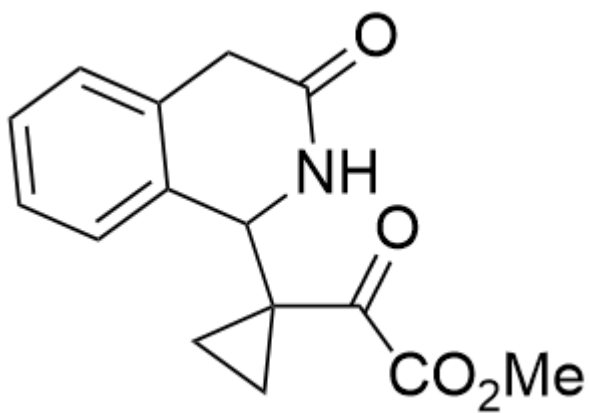
O=C(N1C(C2=CC=CC=C2CC1)=C3CCCl)C3=O

F.



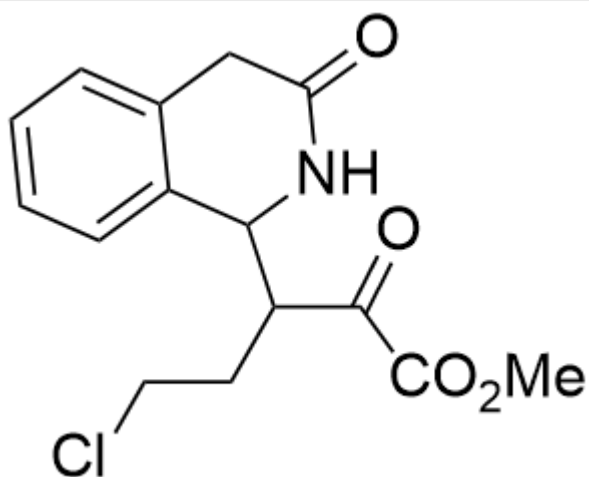
O=C1C2=CC=CC=C2CCN1/C(C(OC)=O)=C3CC\3

G.



O=C1NC(C2(CC2)C(C(OC)=O)=O)C3=CC=CC=C3C1

H.



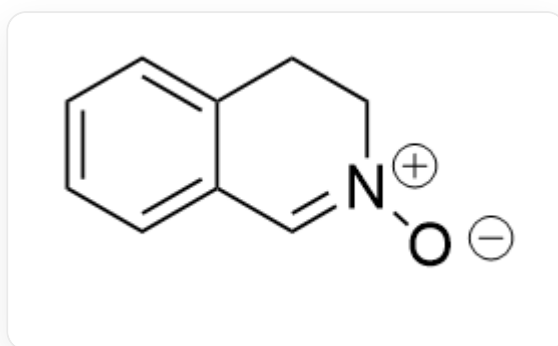
O=C1NC(C(CCCl)C(C(OC)=O)=O)C2=CC=CC=C2C1

Answer

Correct Answer: **E**

Detailed Explanation

According to the question prompt, the first step is a pericyclic reaction of the substrate, inferring that a reverse [3+2] reaction should have occurred to obtain intermediate **1** and intermediate **2**

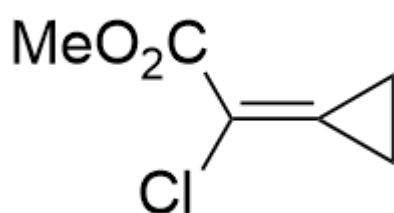


intermediate **1** : [O-]N1[CH+]C2=CC=CC=C2CC1

CHECKPOINT

1 PTS

intermediate **1** : [O-]N1[CH+]C2=CC=CC=C2CC1



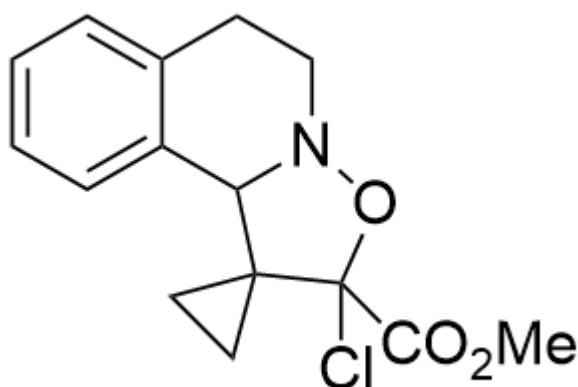
intermediate **2** : Cl/C(C(OC)=O)=C1CC\1

CHECKPOINT

1 PTS

intermediate **2** : Cl/C(C(OC)=O)=C1CC\1

Then, another [3+2] reaction occurs again to form a ring again, yielding intermediate **3**



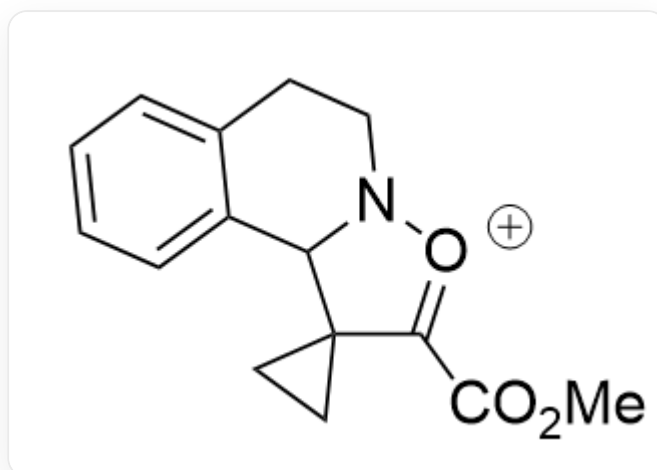
intermediate **3** : ClC1(C(OC)=O)C2(CC2)C3C4=CC=CC=C4CCN3O1

CHECKPOINT

1 PTS

intermediate **3** : ClC1(C(OC)=O)C2(CC2)C3C4=CC=CC=C4CCN3O1

At this point, Cl^- easily departs to form intermediate **4**



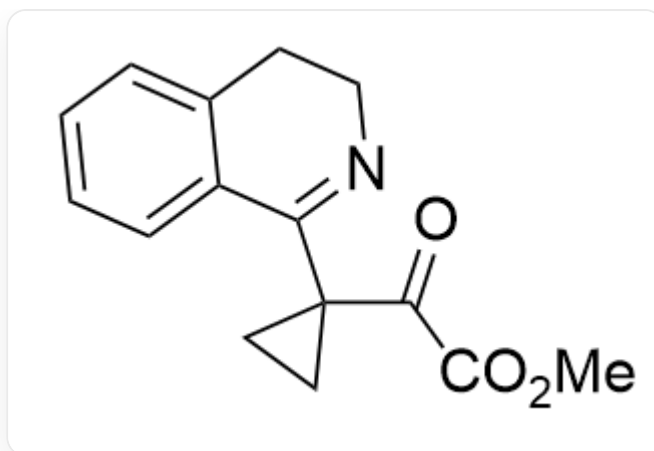
intermediate **4** : O=C(C1=[O+]N2C(C13CC3)C4=CC=CC=C4CC2)OC

CHECKPOINT

1 PTS

intermediate **4** : O=C(C1=[O+]N2C(C13CC3)C4=CC=CC=C4CC2)OC

Subsequently, an elimination reaction occurs to give intermediate **5**



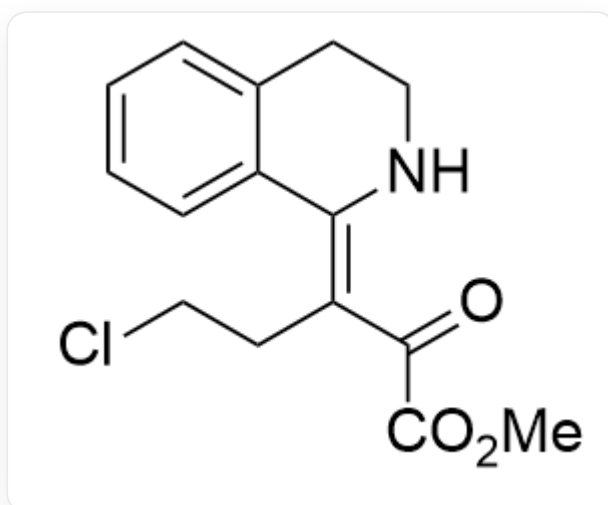
intermediate5 : O=C(C(OC)=O)C1(CC1)C2=NCCC3=CC=CC=C32

CHECKPOINT

1 PTS

intermediate 5 : O=C(C(OC)=O)C1(CC1)C2=NCCC3=CC=CC=C32

At this point, the three-membered ring is subject to electron-withdrawing inductive effects from two electron-withdrawing groups and is easily opened by nucleophilic attack by Cl^- to give intermediate 6



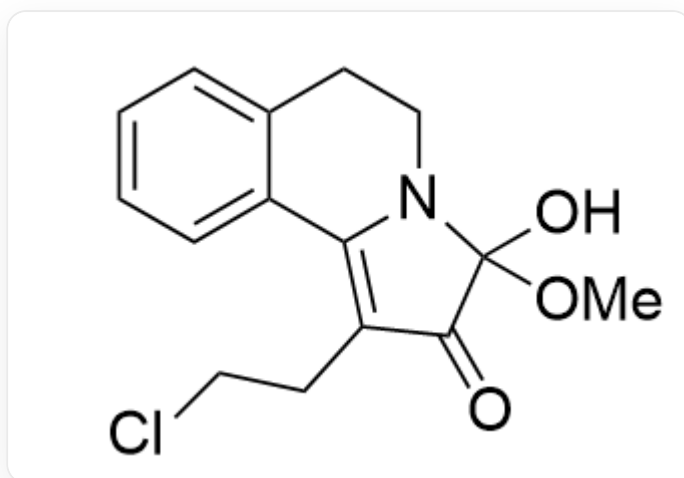
intermediate6 : O=C(C(OC)=O)/C(CCCl)=C1C2=CC=CC=C2CCN1

CHECKPOINT

1 PTS

intermediate **6** : O=C(C(OC)=O)/C(CCCl)=C1C2=CC=CC=C2CCN/1

According to the question stem, the product **A** contains an amide bond, so the next step is an intramolecular nucleophilic substitution reaction to obtain intermediate **7**



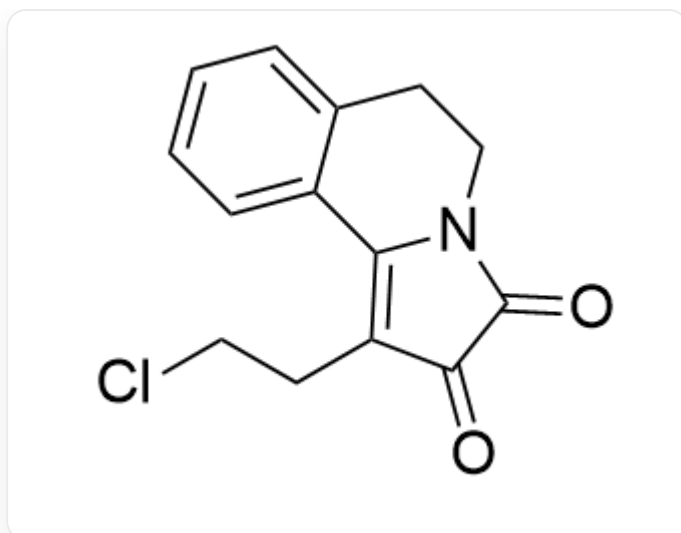
intermediate **7** : O=C1C(CCCl)=C2C3=CC=CC=C3CCN2C1(OC)O

CHECKPOINT

1 PTS

intermediate **7** : O=C1C(CCCl)=C2C3=CC=CC=C3CCN2C1(OC)O

Then, a molecule of methanol is eliminated to obtain the final product **A** containing an amide bond



product **A** : O=C(N1C(C2=CC=CC=C2CC1)=C3CCCI)C3=O